

# 1

---

## *Uncertainty-Related Research Issues in Spatial Analysis*

---

**Daniel A. Griffith,<sup>1</sup> David W. Wong,<sup>2</sup> and Yongwan Chun<sup>1</sup>**

<sup>1</sup>*Geospatial Information Science, School of Economic, Political, and Policy Sciences, University of Texas at Dallas, Dallas, Texas*

<sup>2</sup>*Department of Geography, University of Hong Kong, Pokfulam, Hong Kong, and Department of Geography and Geoinformation Science, George Mason University, Fairfax, Virginia*

### **CONTENTS**

|     |                                                             |    |
|-----|-------------------------------------------------------------|----|
| 1.1 | Introduction.....                                           | 3  |
| 1.2 | Visualization of Uncertainty .....                          | 4  |
| 1.3 | Impacts of Uncertainty on Spatial Patterns and Models ..... | 5  |
| 1.4 | Areal Unit Definitions and Spatial Data Aggregation .....   | 6  |
| 1.5 | Spatial Metadata for Data Quality.....                      | 8  |
| 1.6 | Conclusions.....                                            | 9  |
|     | Acknowledgment.....                                         | 9  |
|     | References.....                                             | 10 |

**ABSTRACT** This chapter enumerates different sources of error in spatial data. It highlights some recent cartographic developments in handling sampling error. We argue that uncertainty in spatial data needs to be addressed because error in spatial data affects the truthfulness of spatial patterns revealed by maps and the effectiveness of developing spatial models. We mention difficulties in defining neighborhoods and addressing aggregation error. Finally, we call for the inclusion of data quality information for each geographical feature in a spatial database as a prerequisite to addressing spatial data quality issues.

---

### **1.1 Introduction**

Spatial data comprise two components: attribute and location information. Attribute information describes the nonlocational characteristics of features, whereas locational information indicates relative or absolute positioning of

these features. Like aspatial data, attribute data have parameters that can be estimated with samples. Thus, one major source of uncertainty is sampling error (i.e., deviations of sample statistics from their corresponding population parameter values). Most often, the scoring of attributes also contains measurement error (i.e., differences between pairs of true and measured values). This additional major source of uncertainty involves proximity of an instrument reading to its corresponding real-world value and includes rounding of numbers and sometimes recording mistakes. Because all models are simplified descriptions of reality—Box (1979, p. 2) said, “All models are wrong, but some are useful”—these descriptions contain specification error (i.e., differences between reality and a model’s representation of it); one goal of science is to minimize this error so that it is not too serious. Only approximate, rather than absolute, positions of features can be tagged to a coordinate system, resulting in location data also having error (i.e., deviations between true and approximate positions), introducing a fourth major source of error for georeferenced data. All four of these sources of error interact, impacting the quality of spatial data and spatial analyses, including mapping. In addition, features or geographical units can be merged or aggregated, perhaps due to confidentiality, data management, or representational concerns. Spatial aggregation can exacerbate and propagate error in spatial data from all four of these sources. Maps containing these errors are more uncertain—their representation of reality may be unnecessarily flawed.

---

## 1.2 Visualization of Uncertainty

Uncertainty in spatial data is a well-researched topic. When the U.S. National Center for Geographic Information and Analysis (NCGIA) formulated its research agenda about this topic in 1988, two initiatives were related to spatial data quality: accuracy of spatial databases and visualizing the quality of spatial information. The first initiative emphasized the sources and management of spatial data uncertainty (Goodchild and Gopal, 1989), whereas the second initiative focused on the visualization of spatial data error (Beard et al., 1991).

Researchers addressing the second NCGIA theme proposed many mapping designs and visualization methods to map data quality information (e.g., Kardos et al., 2005; Leitner and Battenfield, 2000; MacEachren et al., 2005). Although many aspects of the uncertainty of given recorded observed values stemming from sampling, measurement, and locational and specification error can be represented reasonably well with existing methods, few of these methods have been adopted as standard practice or implemented in popular geographic information science (GIS) software. Reasons for including error information on maps have not been obvious to most users

or readers. Map readers often look for geographical patterns, which emerge when the geographical distribution of a variable exhibits a systematic spatial arrangement. But, if observed values for a variable across locations are not statistically different, then the recognized pattern may not be true. Therefore, probably the most pragmatic reason to account for data error when mapping is to be able to determine whether differences between recorded observed values are significant.

Previous methods to map error fall short of assisting readers to differentiate between or compare recorded observed values. The *Atlas of United States Mortality* (Pickle et al., 1996) initiated an attempt to help readers discern attribute uncertainty across locations by including error bars. Sun and Wong (2010) evaluated different approaches to incorporate data reliability information in a choropleth map in the context of mapping estimates from the U.S. American Community Survey (ACS). Adopting the bivariate legend as the preferred design to incorporate error information in choropleth maps, they developed an ArcGIS extension that automates some of the steps in creating bivariate legend choropleth maps using ACS data, or any data, accompanied by the margin of error (MOE) information (Wong and Sun, 2013). The extension also includes several functions for map readers to compare values on a map, determining whether different reported observed map values are statistically different. Such functions are useful to help discern whether patterns that emerge from a set of reported observed map values are artifacts of errors. The extension is freely available to the public. However, more complicated comparisons of recorded observed values—ones that account for different types of error—are still difficult.

Recently, a new classification method to determine class breaks in choropleth maps was proposed (Sun et al., 2014). The class separability method considers error of the estimates in determining classes such that estimates in different classes are statistically different to a large extent. While this classification method may produce maps with relatively unbalanced classes, the spatial patterns revealed by the classification method are more reliable than those using other traditional classification methods, such as quantiles.

---

### 1.3 Impacts of Uncertainty on Spatial Patterns and Models

Although map readers search for spatial patterns on a map of observed values, more quantitatively oriented analysts prefer to quantify spatial patterns, often in terms of the level of spatial autocorrelation. Using similar concepts, cluster detection techniques often are used to identify attribute hot spots and cold spots. These spatial pattern analysis techniques assume that spatial data are relatively accurate. If the magnitudes of errors in recorded

observed values from the various sources are available, they should be considered in evaluating the presence of spatial autocorrelation or spatial clusters (e.g., constructed confidence intervals should reflect them). Unfortunately, existing practices in evaluating spatial patterns, using either global or local statistics, completely ignore the presence of errors in spatial data, or assume that the data are relatively accurate, such that decisions made based upon the data are not compromised. These practices and assumptions definitely are not warranted (Citro and Kalton, 2007).

After evaluating the spatial patterns of response variable outcomes or events, the next logical step in a spatial analysis is to describe a relationship using aspatial and spatial models. Similar to the evaluation of spatial patterns, these models assume trivial levels of errors in both their predicted values (i.e., specification error) and their explanatory variables (e.g., measurement error). If variables have substantial levels of errors, a modelling effort should include the error information, and specification of the models and their predictions should be adjusted. Kriging furnishes one example of how this should be done, with its prediction map accompanied by its prediction error map (uncertainty), as well as its nugget effect (measurement or specification error) estimate. Often spatial models are formulated in the context of explanatory variables that are neighborhood characteristics, some of which may be derived by summarizing individual attribute outcome characteristics within neighborhoods. The majority of studies to date use existing artificial areal units (e.g., pixels of a given size and census tracts or blocks) as proxies for neighborhoods. But individuals residing within these areal units may not necessarily be influenced by the neighborhood defined by their artificial boundaries. Neighborhoods may extend into nearby units. Thus, the neighborhood characteristics measured for artificial units fail to accurately and precisely reflect real-world neighborhood characteristics.

---

## **1.4 Areal Unit Definitions and Spatial Data Aggregation**

Accessing the effects of neighborhood characteristics on various attribute variable outcomes has been one of the major themes in spatial research (e.g., Kawachi and Berkman, 2003). Many studies adopt a simplistic and pragmatic approach to select existing areal units, such as census tracts or block groups, as proxies for neighborhoods. Thus, individuals geocoded to a unit, regardless of their socioeconomic and demographic differences, and their dispersions of locations within the unit, are assumed to have the same neighborhood demarcated by an artificial unit boundary. Apparently, definitions of neighborhoods are extremely complicated, and such simplistic one-size-fits-all approaches to determine neighborhood boundaries for people residing within the same statistical enumeration units work quite

poorly in demarcating individual neighborhoods (e.g., Coulton et al., 2001; Diez-Roux, 2003). Such general practice introduces error in defining neighborhoods for individuals. Unfortunately, often subsequent analyses use such potential erroneous neighborhood definitions to explain attribute variable outcomes. Frequently, physical and socioeconomic characteristics of neighborhoods are used as covariates to explain attribute variable outcomes of individuals within neighborhoods, but the extent to which the individuals are associated with characteristics of a poorly defined neighborhood is completely ignored. Studies adopting the standard frameworks in neighborhood analysis in many societal aspects, such as public health outcomes and crime, potentially commit neighborhood assignment errors, failing to capture the neighborhood information accurately, and thus potentially produce misleading results and decisions.

The modifiable areal unit problem (MAUP), which refers to the inconsistency of analytical results arising from using spatial data tabulated for different resolution levels (scale effect) and for different zonal configurations of similar unit numbers (zoning effect) (Wong, 2009), is a recurring concern in the first theme of the NCGIA initiative (accuracy of spatial databases). Although no general solutions to the MAUP exist, such inconsistencies in results may be regarded as uncertainty associated with spatial data and have significant implications for and impacts on spatial studies (e.g., Grubestic and Matisziw, 2006; Wieczorek et al., 2012). Of the two subproblems constituting the MAUP, the zoning problem may be partially handled with spatial interpolation (Fisher and Langford, 1995), whereas the scale problem seems to defy any meaningful correction or adjustment. Luo et al. (2010) examine the impact of spatial aggregation on a statistical analysis of late-stage breast cancer. They show that aggregation changes data distributions of variables, and subsequently, statistical results are sensitive to a spatial aggregation level.

Griffith et al. (1998) conducted analyses at the individual, census block, census block group, and census tract levels and report that regardless of geographic resolution, pediatric blood lead levels contain positive spatial autocorrelation, with the same conspicuous spatial patterns and with selected covariate relationships that are stable across geographic resolutions. Griffith et al. (2007) discuss the properties of positional error in a street address matching exercise and the allocation of point locations to census geography units. Wong (2009) has written about the MAUP in general, its impacts on multivariate statistical analysis (Fotheringham and Wong, 1991), measuring segregation (Wong, 1997; Wong et al., 1999), and clustering patterns (Wieczorek et al., 2012). He also has introduced a correlation method that is less sensitive to the scale effect (Wong, 2001). Although many researchers treat aggregation error as one aspect of spatial data quality, most work on visualizing spatial data quality focuses only on sampling error, ignoring aggregation error (e.g., Leitner and Buttenfield, 2000; MacEachren et al., 2005; Pickle et al., 1996; Sun and Wong, 2010). Therefore, documenting and

visualizing aggregation error from spatial data for multiple geographical resolutions are desperately needed.

In general, finer-resolution data (i.e., lower-aggregation level) use smaller areal units, and therefore have less of a scale effect and vice versa. Therefore, data tabulated for smaller areal units are preferable because they have less aggregation error. However, data for smaller areal units frequently are based on smaller samples, and therefore their sampling error levels become substantially greater. Aggregating smaller areal units to form larger units tends to yield more stable recorded observed values, such as disease rates. Bolstering such recorded observed values by borrowing information from nearby areal units also tends to stabilize them, but at a cost of additional error propagation or lowering the spatial resolution of the data. Thus, the role of aggregation error and its relationship with sampling error need to be investigated more thoroughly when highly aggregated data are the ones often used in practice.

---

## 1.5 Spatial Metadata for Data Quality

Wong and Wu (1996) introduce the concept of spatial metadata, data about the quality of each attribute for each feature in a GIS context. Recently, this spatial metadata concept has been adopted for such purposes as dissemination of the U.S. ACS and the U.S. Current Population Survey (CPS) data, where the distributed databases include the MOE for each estimate. However, such data quality information is limited to sampling error. The feature-based spatial metadata notion has been adopted by several prototype systems in managing feature-based metadata (e.g., Devillers et al., 2005; Gan and Shi, 2002; Qiu and Hunter, 2002). These systems are relatively complicated and deal with most data quality aspects listed in the U.S. Federal Geographic Data Committee (FGDC) metadata standards (FGDC, 1998). Systems with capabilities to manage and document spatial aggregation error, especially when the spatial data sets are for larger geographical entities comprising smaller areal units, essentially do not exist, with the exception of the Data Uncertainty Engine (DUE) (Brown and Heuvelink, 2007). DUE data quality information is derived through simulation experiments based upon assumptions about certain error sources and properties. For example, the DUE focuses on measurement and locational error and all but ignores sampling and specification error. Also, these types of systems are very sophisticated and involve complicated operations that often are beyond the comprehension of most data users, cartographers, GIS professionals, and some researchers.

Sometimes spatial data sets include uncertainty indices to accompany reported observed map values. For example, the ACS and the CPS include MOEs with their recorded values. Therefore, sampling error information

for these georeferenced data is readily available for use in GIS work (Wong and Sun, 2013). When data at multiple geographical levels are available, some form of aggregation error can be derived empirically. Users at all levels also need to be encouraged to consider errors contained in spatial data in a routine and serious fashion, because the ramifications of not considering errors may be significant (Citro and Kalton, 2007). As pointed out by Heuvelink and Burrough (2002, p. 111), “[it] is crucially important to know how accurate the data contained in spatial databases really are, because without that knowledge we cannot assess the true value of the derived information, nor the correctness of the decisions it supports.” Unfortunately, existing environments for handling spatial data are ill-equipped with tools to derive and manage aggregation error information, and the situation needs to be improved.

---

## **1.6 Conclusions**

In conclusion, error in spatial data originates from multiple sources, and different types of error often tend to interact. Although disentangling these various interactions may be nearly impossible, if not impossible, understanding the nature of different sources of error and their interactions is the first step to advance this line of research. Spatial analysts need to continue to develop more sophisticated modelling procedures and tools to facilitate the handling of error during the use of spatial data, whether within a GIS, through mapping, or during spatial analysis. In parallel, these researchers need to continue to educate both the academic and general public, as well as all categories of data users, about “all [spatial] data hav[ing] error.” And, they need to pursue a research agenda that addresses the visualization of uncertainty, impacts of uncertainty on spatial patterns and models, relationships between areal unit definitions and error propagation via spatial data aggregation, and general data quality, particularly in terms of metadata.

---

## **Acknowledgment**

Research reported in this chapter was supported by the Eunice Kennedy Shriver National Institute of Child Health and Human Development of the National Institutes of Health under award R01HD076020. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

---

## References

- Beard, M.K., Battenfield, B.P., and Clapham, S.B. NCGIA research initiative 7: Visualization of spatial data quality. NCGIA Technical Paper 91-26. University of California, Santa Barbara National Center for Geographic Information and Analysis, 1991.
- Box, G. Robustness in the strategy of scientific model building. In R. Launer and G. Wilderson (eds.), *Robustness in Statistics: Proceedings of a Workshop*. Academic Press, New York, 1979, 201–236.
- Brown, J.D., and Heuvelink, G. The Data Uncertainty Engine (DUE): A software tool for assessing and simulating uncertain environmental variables. *Computers and Geosciences*, 33(2), 172–190, 2007.
- Citro, C.F., and Kalton, G. (eds.). *Using the American Community Survey: Benefits and Challenges*. National Academies Press, Washington, DC, 2007.
- Coulton, C.J., Korbin, J., Chan, T., and Su, M. Mapping residents' perceptions of neighborhood boundaries: Methodological note. *American Journal of Community Psychology*, 29, 371–383, 2001.
- Devillers, R., Bédard, Y., and Jeansoulin, R. Multidimensional management of geospatial data quality information for its dynamic use within GIS. *Photogrammetric Engineering and Remote Sensing*, 71, 205–15, 2005.
- Diez-Roux, A.V. The examination of neighborhood effects on health: Conceptual and methodological issues related to the presence of multiple levels of organization. In I. Kawachi and L.F. Berkman (eds.), *Neighborhoods and Health*. Oxford University Press, Oxford, 2003, 45–64.
- Federal Geographic Data Committee (FGDC). Content standard for digital geospatial metadata. FGDC-STD-001-1998. FGDC, Washington, DC, USA, 1998.
- Fisher, P.F., and Langford, M. Modeling the errors in areal interpolation between zonal systems by Monte Carlo simulation. *Environment and Planning A*, 27(2), 211–224, 1995.
- Fotheringham, A.S., and Wong, D. The modifiable areal unit problem in multivariate statistical analysis. *Environment and Planning A*, 23, 1025–1044, 1991.
- Gan, E., and Shi, W. Error metadata management system. In W. Shi, P.F. Fisher, and M.F. Goodchild (eds.), *Spatial Data Quality*. Taylor & Francis, London, 2002, 251–66.
- Goodchild, M.F., and Gopal, S. (eds.). *Accuracy of Spatial Databases*. Taylor & Francis, London, 1989.
- Griffith, D., Doyle, P., Wheeler, D., and Johnson, D. A tale of two swaths: Urban childhood blood lead levels across Syracuse, NY. *Annals of the Association of American Geographers*, 88, 640–665, 1998.
- Griffith, D., Millones, M., Vincent, M., Johnson, D., and Hunt, A. Impacts of positional error on spatial regression analysis: A case study of address locations in Syracuse, NY. *Transactions in GIS*, 11, 655–679, 2007.
- Grubestic, T.H., and Matisziw, T.C. On the use of ZIP codes and ZIP code tabulation areas (ZCTAs) for the spatial analysis of epidemiological data. *International Journal of Health Geographics*, 5, 58, 2006.



- Heuvelink, G.B.M., and Burrough, P.A. Developments in statistical approaches to spatial uncertainty and its propagation. *International Journal of Geographical Information Science*, 16(2), 111–113, 2002.
- Kardos, J., Benwell, G.L., and Moore, A. The visualization of uncertainty for spatially referenced census data using hierarchical tessellations. *Transactions in GIS*, 9(1), 19–34, 2005.
- Kawachi, I., and Berkman, L.F. *Neighborhoods and Health*. Oxford University Press, Oxford, 2003.
- Leitner, M., and Buttenfield, B.P. Guidelines for the display of attribute certainty. *Cartography and Geographic Information Science*, 27(1), 3–14, 2000.
- Luo, L., McLafferty, S., and Wang, F. Analyzing spatial aggregation error in statistical models of late-stage cancer risk: A Monte Carlo simulation approach. *International Journal of Health Geographics*, 9, 51, 2010.
- MacEachren, A.M., Robinson, A., Hopper, S., Gardner, S., Murray, R., Gahegan, M., and Hetzler, E. Visualizing geospatial information uncertainty: What we know and what we need to know. *Cartography and Geographic Information Science*, 32(3), 139–160, 2005.
- Pickle, L.W., Mungiole, M., Jones, G.K., and White, A.A. *Atlas of United States Mortality*. National Center for Health Statistics, Hyattsville, MD, 1996.
- Qiu, J., and Hunter, G.J. A GIS with the capacity for managing data quality information. In W. Shi, P.F. Fisher, and M.F. Goodchild (eds.), *Spatial Data Quality*. Taylor & Francis, London, 2002, 230–250.
- Sun, M., and Wong, D.W.S. Incorporating data quality information in mapping the American Community Survey data. *Cartography and Geographic Information Science*, 37(4), 285–300, 2010.
- Sun, M., Wong, D.W., and Kronenfeld, B.J. A classification method for choropleth maps incorporating data reliability information. *Professional Geographer*, 2014. DOI: 10.1080/00330124.2014.888627.
- Wieczorek, W.F., Delmerico, A.M., Rogerson, P.A., and Wong, D.W.S. Clusters in irregular areas and lattices. *Wiley Interdisciplinary Reviews (WIREs): Computational Statistics*, 4(1), 67–74, 2012.
- Wong, D. The modifiable areal unit problem (MAUP). In A.S. Fotheringham and P.A. Rogerson (eds.), *The SAGE Handbook of Spatial Analysis*. Sage, London, 2009, 105–123.
- Wong, D.W., and Sun, M. Handling data quality information of survey data in GIS: A case of using the American Community Survey data. *Spatial Demography*, 1(1), 3–16, 2013.
- Wong, D.W.S. Spatial dependency of segregation indices. *Canadian Geographer*, 41(2), 128–136, 1997.
- Wong, D.W.S. Location-specific cumulative distribution function (LSCDF): An alternative to spatial correlation analysis. *Geographical Analysis*, 33(1), 76–93, 2001.
- Wong, D.W.S., Lasus, H., and Falk, R.F. Exploring the variability of segregation index D with scale and zonal systems: An analysis of thirty U.S. cities. *Environment and Planning A*, 31, 507–522, 1999.
- Wong, D.W.S., and Wu, C.V. Spatial metadata and GIS for decision support. In J.F. Nunamaker Jr. and R.H. Sprague Jr. (eds.), *Proceedings of the 29th Hawaii International Conference on System Science (HICSS)*, Vol. 3: *Collaboration Systems and Technology*. IEEE Computer Society Press, Washington, DC, 1996, 557–566.